

PIPES AND CISTERNS

Type-1

1. A can fill a tank in 20 min. and 'B' can empty in 30 min. If these two pipes are opened at the same time, then the tank will be filled or emptied in how much time?

1) filled in 60 min 2) emptied in 60 min 3) filled in 40 min 4) filled in 30 min

2. A tank has two pipes. The first pipe can fill it in 4 hours and the second can empty it in 16 hours. If two pipes be opened together at a time then the tank will be filled in

(a) $5\frac{1}{2}$ hours (b) 6 hours (c) 10 hours (d) $5\frac{1}{3}$ hours

3. A pipe can empty a tank in 15 hrs and another pipe can empty it in 10 hours. If both the pipes are opened simultaneously. Find the time in which a full tank is emptied.

(a) 8 hrs (b) 6 hrs. (c) 4 hrs. (d) 5 hrs.

4. Two pipes A and B can fill a tank with water in 30 minutes and 45 minutes respectively. The third pipe C can empty the tank in 36 minutes. First A and B are opened. After 12 minutes C is opened. Total time (in minutes) in which the tank will be filled up is:

(a) 12 (b) 24 (c) 30 (d) 36

5. Two inlets 'A' and 'B' can fill a tank in 10 min and 15 min, the outlet 'C' can empty the tank in 30 min. If these three pipes are opened at the same time, then the tank will be filled in how much time?

1) 15 min 2) $7\frac{1}{2}$ min 3) 10 min 4) $6\frac{1}{2}$ min

6. Two inlets 'A' and 'B' can fill a tank in 10 min and 15 min., the outlet 'C' can empty the tank in 5 min. If these three pipes are opened at the same time, then the tank will be filled or emptied in how much time?

1) empty in 30 min 2) fill in 10 min 3) empty in 20 min 4) fill in 20 min

Type-2

7. Pipe 'A' can fill a tank in 6 hrs because of a leak at the bottom of the tank, it takes 3 hrs more to fill the tank. Then find the capacity of outlet?

1) 16 hrs 2) 14 hrs 3) 18 hrs 4) 20 hrs

8. Two taps A and B can fill a tank in 10 hours and 12 hours respectively. There is an outlet tap C. If all the taps are opened together the tank will fill in 30 hours. In how many hours tap C can alone empty the tank.

(a) $60/7$ hours (b) $60/9$ hours (c) $60/11$ hours (d) $60/13$ hours

9. Three taps A, B and C together can fill an empty cistern in 10 minutes. The tap A alone can fill it in 30 minutes and the tap B alone in 40 minutes. How long will the tap C alone take to fill it?

(a) 16 minutes (b) 24 minutes (c) 32 minutes (d) 40 minutes

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Type-3

10. Two inlets 'A' and 'B' can fill a tank in 20 min and 30 min, the outlet 'C' can empty the tank in 15 min. First 'A' opens, after 4 min. B opens, after 10 min. from the start 'C' also opens, then the tank will be filled or emptied in how much time?

1) empty in 28 min 2) fill in 24 min 3) fill in 28 min 4) empty in 24 min

11. Pipe A can fill a tank in 12 hours and pipe B can fill it in 15 hours, separately. A third pipe C can empty it in 20 hours. Initially pipe A was opened, after one hour pipe B was opened and then after 1 hour when pipe B was opened pipe C was also opened. In how many hours the tank will be full?

(a) $9\frac{2}{3}$ hours (b) $6\frac{2}{3}$ hours (c) 10 hours (d) None of these

12. Three pipes A, B, and C are attached to a cistern. A can fill it in 10 minutes and B in 15 minutes. C is a waste pipe for emptying it. After opening both the pipes A and B, a man leaves the cistern and returns when the cistern should have been just full. Finding however, that the waste pipe has been left open, he closes it and the cistern is now full in 2 minutes. In how much time the pipe C, if opened alone, empty the full cistern.

(a) 18 min. (b) 16 min. (c) 12 min. (d) None of these

Type-4

13. 12 taps are fitted in a tank some are inlet taps and some are outlet taps. Each inlet tap can fill the tank in 6 hours and each outlet tap can empty the tank in 12 hours. If all the taps are open together then the tank will be full in 4 hours. Then find the number of inlet taps.

(a) 5 (b) 4 (c) 6 (d) None of these

14. An inlet pipe can fill a tank in 5 hours and an outlet pipe can empty the same tank in 36 hours, working individually. How many additional number of outlet pipes of the same capacity are required to be opened, so that the tank never overflows ?

(a) 3 (b) 6 (c) 8 (d) 7

15. 8 taps are fitted in a tank some are inlet taps and some are outlet tap. Each inlet tap can fill the tank in 12 hours and each outlet tap can empty it in 36 hours. Then calculate the number of inlet water taps if the whole tank filled in 3 hours.

(a) 5 (b) 3 (c) 4 (d) None

Type-5

16. A leak in the bottom of a tank can empty it in 6 hours. A tap fills the tank at the rate of 4 liters/min is turned on. If both the taps are opened then the tank will empty in 8 hours. Find the capacity of the tank?

(a) 2400 liters (b) 5780 liters (c) 5760 liters (d) None of these

17. A leak in the bottom of a tank can empty it in 12 hours. A tap which can fill 20 liters of water per minute is turned on. Both the taps are opened now, then the tank is emptied in 20 hours. Find the capacity of the tank ?

(a) 36000 liters (b) 3600 liters (c) 360 liters (d) None of these

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18. Three pipes A, B and C are attached to a cistern pipe A and B can fill it in 30 hours and 20 hours respectively, and third pipe C can leak out 45 liters water per minute. If all the three pipes are opened simultaneously the cistern will be filled in 15 hours. Find the capacity of the cistern ?

(a) 162,000 liters (b) 160,00 liters (c) 5760 liters (d) 150,000 liters

Type-6

19. A tank can be filled by pipe A in 2 hours and pipe B in 6 hours. At 10 am pipe A was opened. At what time will the tank be filled if pipe B is opened at 11 A.M. ?

(a) 12.45 A.M. (b) 5 P.M. (c) 11.45 A.M. (d) 12 P.M.

20. 67. A cistern can be filled by three pipes A, B and C in 10,15 and 25 hours respectively. All three pipes are open altogether at 7 am. At what time will the cistern be full?

(a) 10 : 50 am (b) 11 : 50 pm (c) 12 : 00pm (d) 11 : 50 am